NetApp Solution Automation

NetApp Solutions

Nikhil M Kulkarni
March 02, 2021

This PDF was generated from https://docs.netapp.com/us-en/netapp-solutions/automation/deploy_mellanox_awx.html on July 11, 2021. Always check docs.netapp.com for the latest.

Table of Contents

NetApp Solution Automation 1

NetApp Solution Automation

Procedure

To configure Mellanox switches using Ansible playbooks via AWX/Ansible Tower, follow the below procedure:

1. Configure the AWX/Tower parameters by following the guide here
2. Fill the variables below based on the recommendations/suggestions provided. Once all the variables are filled as per requirement, copy the content by clicking 'Copy' button at the top.

```
<style>
div {
position: relative;
}
div button {
position: absolute;
top: 0;
right: 0;
}
button {
  transition-duration: 0.4s;
  background-color: white;
  color: #1563a3;
  border: 2px solid #1563a3;
}
button:hover {
  background-color: #1563a3;
  color: white;
}
#more_vlans {
  display: block;
}
#more_vlans_button {
  display: none;
}
#more_peerlink_interfaces {
  display: block;
}
#more_peerlink_interfaces_button {
  display: none;
}
#more_storage_interfaces {
  display: block;
}
#more_storage_interfaces_button {
  display: none;
```

```
}
#more_storage_mgmt_interfaces {
  display: block;
}
#more_storage_mgmt_interfaces_button {
  display: none;
}
#more_compute_interfaces {
  display: block;
}
#more_compute_interfaces_button {
  display: none;
}
#more_uplink_interfaces {
  display: block;
}
#more_uplink_interfaces_button {
  display: none;
}
</style>
<div class="listingblock"><div class="content"><div><button id="copy-
button" onclick="CopyClassText()">Copy</button></div><pre><code><div
class="CopyMeClass" id="CopyMeID">># vars file for nar_hci_mellanox_deploy

#Hosts group name
#Default hosts group name - 'mellanox'
#Change only if you are changing the hosts group name either in hosts file
or in inventory groups in case of AWX/Tower
hosts: <span <div contenteditable="true"/><i>mellanox</i></span>

#These set of variables will setup the Mellanox switches for NetApp HCI
that uses a 2-cable compute connectivity option.
#Ansible connection variables for mellanox
ansible_connection: network_cli
ansible_network_os: onyx

#
# Primary Variables
#
#Necessary VLANs for Standard NetApp HCI Deployment [native, Management,
iSCSI_Storage, vMotion, VM_Network, IPL]
#Any additional VLANs can be added to this in the prescribed format below
<br>netapp_hci_vlans:
  - {vlan_id: <span <div contenteditable="true"/><i>2</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>OB_Management</i></span>&quot;}
```

```
  - {vlan_id: <span <div contenteditable="true"/><i>3488</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>IB_Managment</i></span>&quot;}
  - {vlan_id: <span <div contenteditable="true"/><i>3489</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>vMotion</i></span>&quot;}
  - {vlan_id: <span <div contenteditable="true"/><i>3490</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>iSCSI_Storage</i></span>&quot;}
  - {vlan_id: <span <div contenteditable="true"/><i>3487</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>Application</i></span>&quot;}
  - {vlan_id: <span <div contenteditable="true"/><i>3491</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>NFS</i></span>&quot;}
  - {vlan_id: <span <div contenteditable="true"/><i>4000</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>IPL</i></span>&quot;}
  - {vlan_id: <span <div contenteditable="true"/><i>2</i></span>,
vlan_name: &quot;<span <div
contenteditable="true"/><i>Native</i></span>&quot;}
<a id="more_vlans" href="javascript:vlandropdown();">More VLANs</a><div
id="select_more_vlans"></div><a id="more_vlans_button"
href="javascript:addvlans();">Enter VLAN details</a><div
id="extra_vlans"></div>
#LACP load balancing algorithm for IP hash method
#Possible options are: 'destination-mac', 'destination-ip', 'destination-
port', 'source-mac', 'source-ip', 'source-port', 'source-destination-mac',
'source-destination-ip', 'source-destination-port'
#This variable takes multiple options in a single go
#For eg: if you want to configure load to be distributed in the port-
channel based on the traffic source and destination IP address and port
number, use 'source-destination-ip source-destination-port'
#By default, Mellanox sets it to source-destination-mac. Enter the values
below only if you intend to configure any other load balancing algorithm
#Make sure the load balancing algorithm that is set here is also
replicated on the host side
#Recommended algorithm is source-destination-ip source-destination-port
lacp_load_balance: &quot;<span <div contenteditable="true"/><i>source-
destination-ip source-destination-port</i></span>&quot;

#----------------------
# IPL variables
#----------------------

#Inter-Peer Link Portchannel
```

```
#ipl_portchannel to be defined in the format - Po100
ipl_portchannel: <span <div contenteditable="true"/><i>Po100</i></span>

#Inter-Peer Link Addresses
#The IPL IP address should not be part of the management network. This is
typically a private network
ipl_ipaddr_a: <span <div contenteditable="true"/><i>10.0.0.1</i></span>
ipl_ipaddr_b: <span <div contenteditable="true"/><i>10.0.0.2</i></span>

#Define the subnet mask in CIDR number format. Eg: For subnet /22, use
ipl_ip_subnet: 22
ipl_ip_subnet: <span <div contenteditable="true"/><i>24</i></span>

#Inter-Peer Link Interfaces
#members to be defined with Eth in the format. Eg: Eth1/1

peer_link_interfaces:
  description: &quot;<span <div contenteditable="true"/><i>peer link
interfaces</i></span>&quot;
  members:
    - &quot;<span <div contenteditable="true"/><i>Eth1/20</i></span>&quot;
    - &quot;<span <div contenteditable="true"/><i>Eth1/22</i></span>&quot;
<a id="more_peerlink_interfaces" href="javascript:ipldropdown();">More
Peer-Link Interfaces</a><div id="select_more_peerlink_interfaces"></div><a
id="more_peerlink_interfaces_button" href="javascript:addipls();">Enter
peer-link Interface details</a><div id="extra_peerlink_interfaces"></div>
#MLAG VIP IP address should be in the same subnet as that of the switches'
mgmt0 interface subnet
#mlag_vip_ip to be defined in the format - <vip_ip>/<subnet_mask>. Eg:
x.x.x.x/y and must be unique for each MLAG domain
mlag_vip_ip: <span <div
contenteditable="true"/><i>10.195.60.111/24</i></span>

#MLAG VIP Domain Name
#The mlag domain must be unique name for each mlag domain.
#In case you have more than one pair of MLAG switches on the same network,
each domain (consist of two switches) should be configured with different
name.
mlag_domain_name: <span <div contenteditable="true"/><i>MLAG-VIP-DOM-10-
16</i></span>

#---------------------
# Interface Details
#---------------------

#Storage Bond10G Interface details
```

```
#members to be defined with Eth in the format. Eg: Eth1/1
#Only numerical digits between 100 to 1000 allowed for mlag_id
#Operational link speed [variable 'speed' below] to be defined in terms of
bytes.
#For 10 Gigabyte operational speed, define 10G. [Possible values - 10G and
25G]
#Interface descriptions append storage node data port numbers assuming all
Storage Nodes' Port C -> Mellanox Switch A and all Storage Nodes' Port D
-> Mellanox Switch B
#List the storage Bond10G interfaces, their description, speed and MLAG
IDs in list of dictionaries format

storage_interfaces:
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/13</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_01</i></span>&quot;, mlag_id:
<span <div contenteditable="true"/><i>113</i></span>, speed: <span <div
contenteditable="true"/><i>10G</i></span>}
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/14</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_02</i></span>&quot;, mlag_id:
<span <div contenteditable="true"/><i>114</i></span>, speed: <span <div
contenteditable="true"/><i>10G</i></span>}
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/15</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_03</i></span>&quot;, mlag_id:
<span <div contenteditable="true"/><i>115</i></span>, speed: <span <div
contenteditable="true"/><i>10G</i></span>}
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/16</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_04</i></span>&quot;, mlag_id:
<span <div contenteditable="true"/><i>116</i></span>, speed: <span <div
contenteditable="true"/><i>10G</i></span>}
<a id="more_storage_interfaces" href="javascript:storagedropdown();">More
Interfaces towards Storage Nodes</a><div
id="select_more_storage_interfaces"></div><a
id="more_storage_interfaces_button"
href="javascript:addstorageinterfaces();">Enter Storage Interface
details</a><div id="extra_storage_interfaces"></div>
#Storage Bond1G Interface
#Mention whether or not these Mellanox switches will also be used for
Storage Node Mgmt connections
```

```
#Possible inputs for storage_mgmt are 'yes' and 'no'
storage_mgmt: <span <div contenteditable="true"/><i>no</i></span>

#Storage Bond1G (Mgmt) interface details. Only if 'storage_mgmt' is set to
'yes'
#Members to be defined with Eth in the format. Eg: Eth1/1
#Interface descriptions append storage node management port numbers
assuming all Storage Nodes' Port A -> Mellanox Switch A and all Storage
Nodes' Port B -> Mellanox Switch B
#List the storage Bond1G interfaces and their description in list of
dictionaries format

storage_mgmt_interfaces:
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/5</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_01</i></span>&quot;}
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/6</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_02</i></span>&quot;}
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/7</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_03</i></span>&quot;}
  - {members: &quot;<span <div
contenteditable="true"/><i>Eth1/8</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_04</i></span>&quot;}
<a id="more_storage_mgmt_interfaces"
href="javascript:storagemgmtdropdown();">More management interfaces
towards storage nodes</a><div
id="select_more_storage_mgmt_interfaces"></div><a
id="more_storage_mgmt_interfaces_button"
href="javascript:addstoragemgmtinterfaces();">Enter Storage Management
Interface details</a><div id="extra_storage_mgmt_interfaces"></div>
#Compute Interface details
#Members to be defined with Eth in the format. Eg: Eth1/1
#Fill the mlag_id field only if you intend to configure interfaces of
compute nodes into bond or LAG with LACP
#In case you do not intend to configure LACP on interfaces of compute
nodes, either leave the mlag_id field unfilled or comment it or enter NA
in the mlag_id field
#In case you have a mixed architecture where some compute nodes require
LACP and some don't,
#1. Fill the mlag_id field with appropriate MLAG ID for interfaces that
```

```
connect to compute nodes requiring LACP
#2. Either fill NA or leave the mlag_id field blank or comment it for
interfaces connecting to compute nodes that do not require LACP
#Only numerical digits between 100 to 1000 allowed for mlag_id.
#Operational link speed [variable 'speed' below] to be defined in terms of
bytes.
#For 10 Gigabyte operational speed, define 10G. [Possible values - 10G and
25G]
#Interface descriptions append compute node port numbers assuming all
Compute Nodes' Port D -> Mellanox Switch A and all Compute Nodes' Port E
-> Mellanox Switch B
#List the compute interfaces, their speed, MLAG IDs and their description
in list of dictionaries format

compute_interfaces:
  - members: &quot;<span <div
contenteditable="true"/><i>Eth1/2</i></span>&quot;
    description: &quot;<span <div
contenteditable="true"/><i>HCI_Compute_Node_01</i></span>&quot;
    mlag_id: <span <div contenteditable="true"/><i> </i></span>  #Fill the
mlag_id only if you wish to use LACP on interfaces towards compute nodes
    speed: <span <div contenteditable="true"/><i>10G</i></span>
  - members: &quot;<span <div
contenteditable="true"/><i>Eth1/4</i></span>&quot;
    description: &quot;<span <div
contenteditable="true"/><i>HCI_Compute_Node_02</i></span>&quot;
    mlag_id: <span <div contenteditable="true"/><i>104</i></span>  #Fill
the mlag_id only if you wish to use LACP on interfaces towards compute
nodes
    speed: <span <div contenteditable="true"/><i>10G</i></span>
<a id="more_compute_interfaces" href="javascript:computedropdown();">More
interfaces towards compute nodes</a><div
id="select_more_compute_interfaces"></div><a
id="more_compute_interfaces_button"
href="javascript:addcomputeinterfaces();">Enter Compute Interface
details</a><div id="extra_compute_interfaces"></div>
#Spanning-tree protocol type for uplink connections.
#The valid options are 'network' and 'normal'; selection depends on the
uplink switch model.
uplink_stp_type: <span <div contenteditable="true"/><i>network</i></span>

#Uplink Switch LACP support
#Possible options are 'yes' and 'no' - Set to 'yes' only if your uplink
switch supports LACP
uplink_switch_lacp: <span <div contenteditable="true"/><i>yes</i></span>
```

```
#Uplink Interface details
#Members to be defined with Eth in the format. Eg: Eth1/1
#Only numerical digits between 100 to 1000 allowed for mlag_id.
#Operational link speed [variable 'speed' below] to be defined in terms of
bytes.
#For 10 Gigabyte operational speed, define 10G. [Possible values in
Mellanox are 1G, 10G and 25G]
#List the uplink interfaces, their description, MLAG IDs and their speed
in list of dictionaries format

uplink_interfaces:
  - members: &quot;<span <div
contenteditable="true"/><i>Eth1/18</i></span>&quot;
    description_switch_a: &quot;<span <div
contenteditable="true"/><i>SwitchA:Ethx/y ->
Uplink_Switch:Ethx/y</i></span>&quot;
    description_switch_b: &quot;<span <div
contenteditable="true"/><i>SwitchB:Ethx/y ->
Uplink_Switch:Ethx/y</i></span>&quot;
    mlag_id: <span <div contenteditable="true"/><i>118</i></span>  #Fill
the mlag_id only if 'uplink_switch_lacp' is set to 'yes'
    speed: <span <div contenteditable="true"/><i>10G</i></span>
    mtu: <span <div contenteditable="true"/><i>1500</i></span>
<a id="more_uplink_interfaces" href="javascript:uplinkdropdown();">More
interfaces towards uplink switches</a><div
id="select_more_uplink_interfaces"></div><a
id="more_uplink_interfaces_button"
href="javascript:adduplinkinterfaces();">Enter Uplink Interface
details</a><div
id="extra_uplink_interfaces"></div></div></code></pre></div></div>
<script>
function CopyClassText(){
    var textToCopy = document.getElementById("CopyMeID");
    var currentRange;
    if(document.getSelection().rangeCount > 0)
    {
        currentRange = document.getSelection().getRangeAt(0);
        window.getSelection().removeRange(currentRange);
    }
    else
    {
        currentRange = false;
    }
    var CopyRange = document.createRange();
    CopyRange.selectNode(textToCopy);
    window.getSelection().addRange(CopyRange);
```

```
    document.getElementById("more_vlans").style.display = "none";
    document.getElementById("more_peerlink_interfaces").style.display =
"none";
    document.getElementById("more_storage_interfaces").style.display =
"none";
    document.getElementById("more_storage_mgmt_interfaces").style.display
= "none";
    document.getElementById("more_compute_interfaces").style.display =
"none";
    document.getElementById("more_uplink_interfaces").style.display =
"none";
    var command = document.execCommand("copy");
      if (command)
      {
          document.getElementById("copy-button").innerHTML = "Copied!";
          setTimeout(revert_copy, 3000);
      }
    window.getSelection().removeRange(CopyRange);
    if(currentRange)
    {
        window.getSelection().addRange(currentRange);
    }
}
function revert_copy() {
      document.getElementById("copy-button").innerHTML = "Copy";
      document.getElementById("more_vlans").style.display = "block";
      document.getElementById("more_peerlink_interfaces").style.display =
"block";
      document.getElementById("more_storage_interfaces").style.display =
"block";

document.getElementById("more_storage_mgmt_interfaces").style.display =
"block";
      document.getElementById("more_compute_interfaces").style.display =
"block";
      document.getElementById("more_uplink_interfaces").style.display =
"block";
}
function vlandropdown() {
    document.getElementById("more_vlans").style.display = "none";
    document.getElementById("more_vlans_button").style.display = "block";
    var x=1;
    var myHTML = '';
    var buildup = '';
    var wrapper = document.getElementById("select_more_vlans");
    while (x < 100) {
```

```
      buildup += '<option value="' + x + '">' + x + '</option>';
      x++;
    }
    myHTML += '<a id="more_vlans_info">How many extra VLANs do you wish to
add?</a><select name="number_of_extra_vlans" id="number_of_extra_vlans">'
+ buildup + '</select>';
    wrapper.innerHTML = myHTML;
}

function addvlans() {
    var y = document.getElementById("number_of_extra_vlans").value;
    var j=0;
    var myHTML = '';
    var wrapper = document.getElementById("extra_vlans");
    while (j < y) {
        j++;
        myHTML += '  - {vlan_id: <span <div contenteditable="true"/><i>
</i></span>, vlan_name: &quot;<span <div contenteditable="true"/><i>
</i></span>&quot;}<br>';
    }
    wrapper.innerHTML = myHTML;
    document.getElementById("select_more_vlans").style.display = "none";
    document.getElementById("more_vlans_button").style.display = "none";
}
function ipldropdown() {
    document.getElementById("more_peerlink_interfaces").style.display =
"none";

document.getElementById("more_peerlink_interfaces_button").style.display =
"block";
    var x=1;
    var myHTML = '';
    var buildup = '';
    var wrapper =
document.getElementById("select_more_peerlink_interfaces");
    while (x < 10) {
        buildup += '<option value="' + x + '">' + x + '</option>';
        x++;
    }
    myHTML += '<a id="more_peerlink_interfaces_info">How many more Peer-
Link Interfaces do you wish to add?</a><select
name="number_of_extra_peerlink_interfaces"
id="number_of_extra_peerlink_interfaces">' + buildup + '</select>';
    wrapper.innerHTML = myHTML;
}
function addipls() {
```

```
    var y =
document.getElementById("number_of_extra_peerlink_interfaces").value;
    var j=0;
    var myHTML = '';
    var wrapper = document.getElementById("extra_peerlink_interfaces");
    while (j < y) {
        j++;
        myHTML += '    - &quot;<span <div contenteditable="true"/><i>
</i></span>&quot;<br>';
    }
    wrapper.innerHTML = myHTML;

document.getElementById("select_more_peerlink_interfaces").style.display =
"none";

document.getElementById("more_peerlink_interfaces_button").style.display =
"none";
}
function storagedropdown() {
    document.getElementById("more_storage_interfaces").style.display =
"none";

document.getElementById("more_storage_interfaces_button").style.display =
"block";
    var x=1;
    var myHTML = '';
    var buildup = '';
    var wrapper =
document.getElementById("select_more_storage_interfaces");
    while (x < 10) {
        buildup += '<option value="' + x + '">' + x + '</option>';
        x++;
    }
    myHTML += '<a id="more_storage_interfaces_info">How many more Storage
Interfaces do you wish to add?</a><select
name="number_of_extra_storage_interfaces"
id="number_of_extra_storage_interfaces">' + buildup + '</select>';
    wrapper.innerHTML = myHTML;
}
function addstorageinterfaces() {
    var y =
document.getElementById("number_of_extra_storage_interfaces").value;
    var j=0;
    var myHTML = '';
    var wrapper = document.getElementById("extra_storage_interfaces");
    while (j < y) {
```

```
        j++;
        myHTML += '  - {members: &quot;<span <div
contenteditable="true"/><i>Ethx/y</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_x</i></span>&quot;, mlag_id:
<span <div contenteditable="true"/><i>xxx</i></span>, speed: <span <div
contenteditable="true"/><i>10G</i></span>}<br>';
    }
    wrapper.innerHTML = myHTML;

document.getElementById("select_more_storage_interfaces").style.display =
"none";

document.getElementById("more_storage_interfaces_button").style.display =
"none";
}
function storagemgmtdropdown() {
    document.getElementById("more_storage_mgmt_interfaces").style.display
= "none";

document.getElementById("more_storage_mgmt_interfaces_button").style.displ
ay = "block";
    var x=1;
    var myHTML = '';
    var buildup = '';
    var wrapper =
document.getElementById("select_more_storage_mgmt_interfaces");
    while (x < 10) {
        buildup += '<option value="' + x + '">' + x + '</option>';
        x++;
    }
    myHTML += '<a id="more_storage_mgmt_interfaces_info">How many more
Storage Management Interfaces do you wish to add?</a><select
name="number_of_extra_storage_mgmt_interfaces"
id="number_of_extra_storage_mgmt_interfaces">' + buildup + '</select>';
    wrapper.innerHTML = myHTML;
}
function addstoragemgmtinterfaces() {
    var y =
document.getElementById("number_of_extra_storage_mgmt_interfaces").value;
    var j=0;
    var myHTML = '';
    var wrapper =
document.getElementById("extra_storage_mgmt_interfaces");
    while (j < y) {
        j++;
```

```
        myHTML += '  - {members: &quot;<span <div
contenteditable="true"/><i>Ethx/y</i></span>&quot;, description:
&quot;<span <div
contenteditable="true"/><i>HCI_Storage_Node_x</i></span>&quot;}<br>';
    }
    wrapper.innerHTML = myHTML;

document.getElementById("select_more_storage_mgmt_interfaces").style.displ
ay = "none";

document.getElementById("more_storage_mgmt_interfaces_button").style.displ
ay = "none";
}
function computedropdown() {
    document.getElementById("more_compute_interfaces").style.display =
"none";

document.getElementById("more_compute_interfaces_button").style.display =
"block";
    var x=1;
    var myHTML = '';
    var buildup = '';
    var wrapper =
document.getElementById("select_more_compute_interfaces");
    while (x < 10) {
        buildup += '<option value="' + x + '">' + x + '</option>';
        x++;
    }
    myHTML += '<a id="more_compute_interfaces_info">How many more Compute
Interfaces do you wish to add?</a><select
name="number_of_extra_compute_interfaces"
id="number_of_extra_compute_interfaces">' + buildup + '</select>';
    wrapper.innerHTML = myHTML;
}
function addcomputeinterfaces() {
    var y =
document.getElementById("number_of_extra_compute_interfaces").value;
    var j=0;
    var myHTML = '';
    var wrapper = document.getElementById("extra_compute_interfaces");
    while (j < y) {
        j++;
        myHTML += '  - members: &quot;<span <div
contenteditable="true"/><i>Ethx/y</i></span>&quot;<br>    description:
&quot;<span <div
contenteditable="true"/><i>HCI_Compute_Node_x</i></span>&quot;<br>
```

```
mlag_id: <span <div contenteditable="true"/><i> </i></span>  #Fill the
mlag_id only if you wish to use LACP on interfaces towards compute
nodes<br>    speed: <span <div
contenteditable="true"/><i>10G</i></span><br>';
    }
    wrapper.innerHTML = myHTML;

document.getElementById("select_more_compute_interfaces").style.display =
"none";

document.getElementById("more_compute_interfaces_button").style.display =
"none";
}
function uplinkdropdown() {
    document.getElementById("more_uplink_interfaces").style.display =
"none";
    document.getElementById("more_uplink_interfaces_button").style.display
= "block";
    var x=1;
    var myHTML = '';
    var buildup = '';
    var wrapper =
document.getElementById("select_more_uplink_interfaces");
    while (x < 10) {
        buildup += '<option value="' + x + '">' + x + '</option>';
        x++;
    }
    myHTML += '<a id="more_uplink_interfaces_info">How many more Uplink
Interfaces do you wish to add?</a><select
name="number_of_extra_uplink_interfaces"
id="number_of_extra_uplink_interfaces">' + buildup + '</select>';
    wrapper.innerHTML = myHTML;
}
function adduplinkinterfaces() {
    var y =
document.getElementById("number_of_extra_uplink_interfaces").value;
    var j=0;
    var myHTML = '';
    var wrapper = document.getElementById("extra_uplink_interfaces");
    while (j < y) {
        j++;
        myHTML += '  - members: &quot;<span <div
contenteditable="true"/><i>Ethx/y</i></span>&quot;<br>
description_switch_a: &quot;<span <div
contenteditable="true"/><i>SwitchA:Ethx/y ->
Uplink_Switch:Ethx/y</i></span>&quot;<br>    description_switch_b:
```

```
&quot;<span <div contenteditable="true"/><i>SwitchB:Ethx/y ->
Uplink_Switch:Ethx/y</i></span>&quot;<br>    mlag_id: <span <div
contenteditable="true"/><i>xxx</i></span>  #Fill the mlag_id only if
uplink_switch_lacp is set to yes<br>    speed: <span <div
contenteditable="true"/><i>10G</i></span><br>    mtu: <span <div
contenteditable="true"/><i>1500</i></span><br>';
    }
    wrapper.innerHTML = myHTML;
    document.getElementById("select_more_uplink_interfaces").style.display
= "none";
    document.getElementById("more_uplink_interfaces_button").style.display
= "none";
}
</script>
```

3. Login to AWX/Tower, go to 'Resources' → 'Templates' and click on 'Launch' button for the appropriate Mellanox job template.
4. Paste the copied content in the variable field that is prompted, click on 'Next' and then click on 'Launch' to run the job template.

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.